

Diagnosing Glaucoma With Spectral-Domain Optical Coherence Tomography Using Deep Learning Classifier

Jinho Lee, MD,*† Young Kook Kim, MD,*† Ki Ho Park, MD, PhD,*†
and Jin Wook Jeoung, MD, PhD*†

Précis: A spectral-domain optical coherence tomography (SD-OCT) based deep learning system detected glaucomatous structural change with high sensitivity and specificity. It outperformed the clinical diagnostic parameters in discriminating glaucomatous eyes from healthy eyes.

Purpose: The purpose of this study was to assess the performance of a deep learning classifier for the detection of glaucomatous change based on SD-OCT.

Methods: Three hundred fifty image sets of ganglion cell-inner plexiform layer (GCIPL) and retinal nerve fiber layer (RNFL) SD-OCT for 86 glaucomatous eyes and 307 SD-OCT image sets of 196 healthy participants were recruited and split into training (197 eyes) and test (85 eyes) datasets based on a patient-wise split. The bottleneck features extracted from the GCIPL thickness map, GCIPL deviation map, RNFL thickness map, and RNFL deviation map were used as predictors for the deep learning classifier. The area under the receiver operating characteristic curve (AUC) was calculated and compared with those of conventional glaucoma diagnostic parameters including SD-OCT thickness profile and standard automated perimetry (SAP) to evaluate the accuracy of discrimination for each algorithm.

Results: In the test dataset, this deep learning system achieved an AUC of 0.990 [95% confidence interval (CI), 0.975-1.000] with a sensitivity of 94.7% and a specificity of 100.0%, which was significantly larger than the AUCs with all of the optical coherence tomography and SAP parameters: 0.949 (95% CI, 0.921-0.976) with average GCIPL thickness ($P=0.006$), 0.938 (95% CI, 0.905-0.971) with average RNFL thickness ($P=0.003$), and 0.889 (0.844-0.934) with mean deviation of SAP ($P<0.001$; DeLong test).

Conclusion: An SD-OCT-based deep learning system can detect glaucomatous structural change with high sensitivity and specificity.

Key Words: glaucoma, deep learning, optical coherence tomography (*J Glaucoma* 2020;29:287-294)

Glaucoma, affecting >70 million people, is the leading cause of visual impairment worldwide.¹⁻³ Early diagnosis

of glaucoma and proper treatment are essential for the preservation of patients' vision.^{4,5} Optical coherence tomography (OCT) is widely utilized for the detection of early glaucomatous structural loss.⁶ Among other modalities, spectral-domain optical coherence tomography (SD-OCT) has been broadly employed for analysis of the ganglion cell-inner plexiform layer (GCIPL) and retinal nerve fiber layer (RNFL).⁷⁻¹⁰ The thickness and deviation maps of GCIPL and RNFL SD-OCT are colored images based on raw thickness data and an age-matched normative database. However, there are diverse thinning patterns in optic neuropathies such as glaucoma, and even healthy eyes can show nonspecific thinning on such image maps. Although most OCT reports provide both numerical data and color-coded classification according to normative databases for the peripapillary RNFL and macular GCIPL sectors, their final interpretation by clinicians could be subjective.

Artificial intelligence is a rapidly expanding field for integration into clinical practice. Most notably, machine learning including deep neural networks (a.k.a. deep learning) is augmenting diagnostic capabilities. Especially, the convolutional neural network is widely used for image recognition as applied in medical imaging. In ophthalmology, deep learning became impossible to ignore after the revolutionary report on diabetic retinopathy screening.¹¹ Meanwhile, artificial intelligence platforms for application to other ophthalmologic diseases including glaucoma¹²⁻¹⁴ also are emerging. In glaucoma, however, most of the previous work has focused on the detection of RNFL defect on fundus photographs or stereoscopic optic disc photographs.^{12,13} In the field of OCT, Muhammad et al¹⁴ utilized a hybrid deep learning method with a single wide-field OCT protocol, achieving excellent diagnostic accuracy. To the best of our knowledge, however, there is no previous report introducing a deep learning model to integrate GCIPL and RNFL OCT maps for discrimination of glaucomatous damage.

The purpose of this study was to evaluate the diagnostic efficacy of a deep learning algorithm for distinguishing of glaucomatous from normal images on thickness and deviation maps of GCIPL and RNFL SD-OCT.

METHODS

Participants

Eyes were chosen from a database of subjects examined for glaucoma at the Glaucoma Clinic, Seoul National University Hospital, between January 2012 and May 2018. This study was approved by the Institutional Review Board of Seoul National University Hospital, and adhered to the tenets of the Declaration of Helsinki. Informed consent was waived due to the retrospective nature of the study and the fully anonymized usage of the database.

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Reprints: Jin Wook Jeoung, MD, PhD, Department of Ophthalmology, Seoul National University Hospital, Seoul National University College of Medicine, 101 Daehak-ro, Jongno-gu, Seoul 03080, Korea (e-mail: neuroprotect@gmail.com).

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Before the study, all of the subjects underwent a complete ophthalmic examination, including visual acuity assessment, refraction, slit-lamp biomicroscopy, gonioscopy, Goldmann applanation tonometry (Haag-Streit, Koniz, Switzerland), and dilated stereoscopic examination of the optic disc. They also underwent digital color stereo disc photography, red-free RNFL photography (Vx-10; Kowa Optomed Inc., Tokyo, Japan), optic nerve head and macular imaging by high-definition optical coherence tomography (HD-OCT) (Cirrus; Carl Zeiss Meditec Inc., Dublin, CA), and standard automated perimetry (SAP) with the central 24-2 threshold test of the Humphrey visual field (HFA II; Humphrey Instruments Inc., Dublin, CA).

For inclusion, subjects were required to have an intraocular pressure (IOP) ≤ 21 mm Hg and a normal open anterior chamber angle. Individuals were excluded from further analysis based on the following criteria: (1) the existence of a secondary cause of glaucomatous optic neuropathy; (2) any retinal diseases that could affect GC IPL; (3) a history of intraocular surgery (except cataract surgery) or retinal laser photocoagulation; (4) any neurological or systemic diseases that could affect retinal or visual-field results. If both eyes of a patient were eligible for the study, 1 eye was randomly selected.

Cirrus HD-OCT Measurement

Optic-disc (optic disc cube 200×200) and macular scans (macular cube 512×128) using HD-OCT software (Cirrus, version 6.0; Carl Zeiss Meditec Inc.) for RNFL and GC IPL thickness measurements, respectively, were carried out. The analyzed GC IPL thickness parameters were the average, minimum, and 6 sectoral thicknesses. The evaluated RNFL parameters were average, 4-quadrant thickness, and thickness at each of the 12 clock-hour positions.

Diagnosis of Glaucoma and Quality Control

Glaucomatous eyes were defined by the presence of characteristic optic disc change on stereo disc photographs or the presence of RNFL defect on red-free fundus images, regardless of the presence or absence of glaucomatous visual-field defects. Eyes with glaucomatous visual-field defects were defined as (1) a cluster of 3 points with probabilities $< 5\%$ in at least 1 hemifield on the pattern deviation map, including at least 1 point with a probability $< 1\%$ or a cluster of 2 points with a probability $< 1\%$; (2) outside of the normal limits on glaucoma hemifield test; or (3) a pattern standard deviation $< 5\%$, as confirmed by at least 2 reliable examinations.

The healthy controls had no IOP elevation history, no glaucomatous optic disc appearance, no RNFL defects, and normal visual-field test results. The appearance of the optic disc on stereoscopic photographs and that of the RNFL on red-free imaging were evaluated by 3 glaucoma specialists (J.L., J.W.J., and K.H.P.) who were masked to all other information on the eyes. If the opinions on the diagnosis of glaucoma differed, the pertinent eye was excluded from the training and test datasets.

To clarify the glaucoma diagnosis and to train the deep learning model appropriately, field definition also was assessed. Adequate field definition was confirmed only when (1) the entire optic disc, macula, and temporal arcades were visible in the image; (2) and there was no retinal shadow caused by small-pupil artifact.¹⁵ The cases presenting inadequate field definition were excluded from further analysis. Similarly, only subjects with high-quality OCT images of

signal strength ≥ 6 and without motion artifacts, involuntary saccade, overly poor centration or algorithm segmentation failure were included in the study.

Image Processing and Development of Deep Learning Algorithm

The OCT images of the subjects were divided into a training set that was used to train the deep neural network and a test set that was used for evaluation of the algorithm. Image pixel values were scaled to values within the range of 0 to 1 and then downsized to a 331×331 matrix. The ratio between the training and test sets was set to 7:3.

Recently, there have been many efforts to automate the design of a deep neural network in the form of the so-called “neural architecture search (NAS),” commonly known as “NASNet.”^{16,17} It has shown excellent diagnostic performance compared with previous human-designed deep neural networks¹⁷ including Inception-v3¹⁸ and ResNet¹⁹; therefore, we adopted this network as our base architecture. We used the NASNet-A (large) architecture originally trained on ImageNet and discarded the final output layer because ImageNet is designed for discrimination of ordinary objects, not SD-OCT images. All of the NASNet layers were frozen and used to extract the bottleneck features of the SD-OCT images.

For each subject, 4 images were used as the input for the NASNet architecture, which extracted 4032 bottleneck features for each of the following: (1) a GC IPL thickness map; (2) a GC IPL deviation map; (3) an RNFL thickness map; (4) an RNFL deviation map.

First, independent deep learning models were trained using only one kind of OCT image (ie, GC IPL deviation map or thickness map). In each model, the bottleneck features of the input image were processed by the softmax classifier to output the probability of having glaucoma. Second, in the model named “ensemble training model” (Fig. 1), all of the bottleneck features of the 4 images were integrated and considered as the training data of the subsequent deep neural network. The deep neural network consisted of 4 hidden layers including 4000, 2000, 1000, and

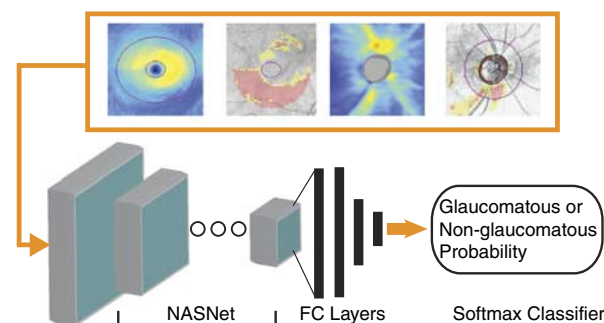


FIGURE 1. Schematic diagram of ensemble training deep learning classifier. A total of 4 SD-OCT images are fed into NASNet, which extracts 4032 bottleneck features for each of the following: (1) GC IPL thickness map; (2) GC IPL deviation map; (3) RNFL thickness map; (4) RNFL deviation map. All of the bottleneck features are integrated and used as input to train the subsequent fully connected layers and final softmax classification layer for the identification of glaucoma probability. FC indicates fully connected; GC IPL, ganglion cell-inner plexiform layer; NAS, neural architecture search; RNFL, retinal nerve fiber layer; SD-OCT, spectral-domain optical coherence tomography. Figure 1 can be viewed in color online at www.glaucomajournal.com.

1000 neurons each and final softmax classification layer. The Adam optimizer (learning rate: 0.001) was introduced for training of the network. To prevent overfitting, dropout (rate: 0.7) and L2 regularization ($\lambda=0.01$) were applied in the training process. The detailed process of the determination of the network size and hyperparameters is described

in the Online Supplemental Text (Supplemental Digital Content 1, <http://links.lww.com/IJG/A360>).

Figure 2 shows representative cases included in this study, including a typical glaucomatous case with inferior disc notching and inferotemporal RNFL defect (Fig. 2A) and a healthy subject (Fig. 2B).

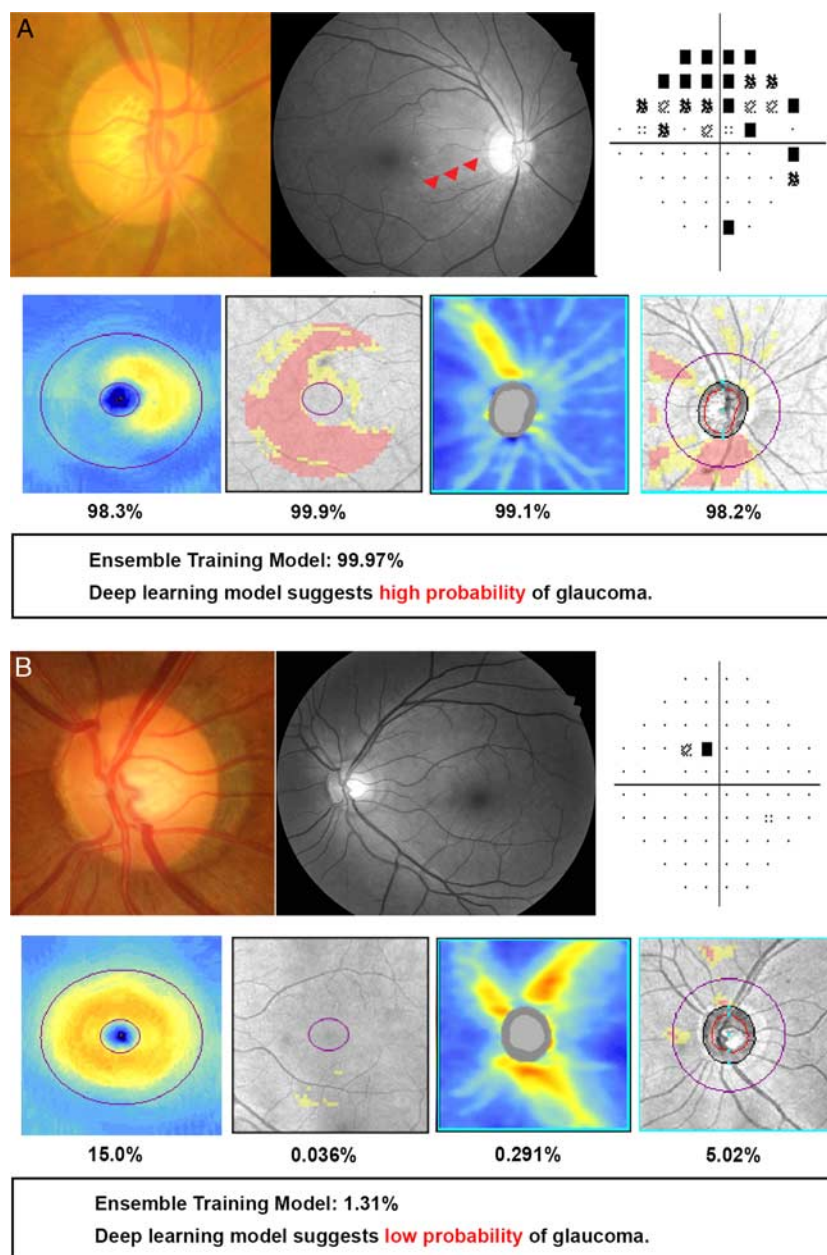


FIGURE 2. Glaucomatous and healthy eyes from the test set. A, Glaucomatous eye of a 74-year-old woman. Inferior notching of the optic disc rim was found on disc photography. A diffuse inferior retinal nerve fiber layer (RNFL) defect corresponding to the superior hemifield defect on the pattern deviation plot was detected by red-free RNFL photography. Glaucoma probability was predicted to 98.3% with the ganglion cell-inner plexiform layer (GCIPL) thickness map, 99.9% with the GCIPL deviation map, 99.1% with the RNFL thickness map, and 98.2% with the RNFL deviation map. The ensemble training model showed a glaucoma probability of 99.97%. B, Healthy eye of a 69-year-old woman. The disc photograph and red-free RNFL photograph showed no definite structural damage. The pattern deviation plot likewise indicated no evidence of glaucomatous damage. Glaucoma probability was predicted to 15.0% with the GCIPL thickness map, 0.036% with the RNFL deviation map, 0.291% with the RNFL thickness map, and 5.02% with the RNFL deviation map. The ensemble training model showed a glaucoma probability of 1.31%, classifying this case as normal. Figure 2 can be viewed in color online at www.glaucomajournal.com.

Since the deep learning model needs as many training samples as possible for achieving high accuracy, if a subject underwent OCT multiple times, all of the images were included for the study. However, for the accurate evaluation of diagnostic efficacy and prevention of model cheating, the whole dataset was split patient-wise into training and test sets.

Statistical Analyses

Between the healthy and glaucoma groups, the numerical values were compared by Student *t* test; sex differences were computed using χ^2 tests. We used the area under the receiver operating characteristic curve (AUC) with 95% confidence intervals (CIs), changing the cutoff value for the probability to be classified as glaucoma.²⁰ DeLong test was used to test for the statistical significance of the AUC between any 2 parameters.²¹ The sensitivities and specificities were calculated according to the optimal cutoff point, which was set as the maximum of the Youden index. All of the statistical analyses were performed using R software (version 3.5.0) for statistics.²² *P*-values <0.05 were considered statistically significant. The data ranges were recorded as mean $\pm$ SD.

RESULTS

Study Subjects and OCT Image Datasets

On the basis of a medical record review, we recruited 86 eyes of 86 glaucoma patients and 196 eyes of 196 healthy controls. Among the glaucomatous eyes, 47 were in the early stage (SAP MD > -6.0 dB), although the other 39 were in the moderate-to-severe stage (SAP MD $\leq$ -6.0 dB). The total number of obtained image sets (GCIPL/RNFL thickness and deviation maps), including multiple acquisitions of OCT images at each outpatient department, were 350 for the glaucoma group and 307 for the healthy controls. Of the total of 282 eyes, 197 were assigned to the training set while 85 were assigned to the test set. The study population's clinical characteristics are provided in Table 1. Sex predilection and refractive error were not significantly different between the 2 groups. However, age and baseline IOP showed significant differences (*P* = 0.005 and 0.009, respectively). The glaucomatous eyes manifested thinner RNFL and GCIPL thicknesses on average in all sectors and worse mean deviation (MD) of SAP (all *P* < 0.001).

Diagnostic Capability of Deep Learning Prediction Model

Table 2 summarizes the AUCs, sensitivities, and specificities of the deep learning classifiers and conventional diagnostic parameters. The ensemble training model AUC for discrimination of glaucomatous changes between normal and glaucomatous cases in the test set (0.990; 95% CI, 0.975-1.000) was excellent. It was a significantly higher AUC than those of any of the OCT thicknesses as well as SAP parameters. In the deep learning models trained with a single modality, only the model trained with the RNFL deviation map showed a significantly lower AUC than that of the ensemble training model (*P* = 0.026), and only the model trained by the GCIPL deviation map showed superior performance to all of the conventional diagnostic parameters (Table 3). Among the GCIPL thickness parameters, minimum GCIPL thickness had the best AUC (0.963; 95% CI, 0.934-0.985). Among the RNFL thickness

TABLE 1. Descriptive Statistics of Participants

	Nonglaucomatous Eyes (196 Eyes of 196 Controls)	Glaucomatous Eyes (86 Eyes of 86 Patients)	<i>P</i>
Age (y)	61.2 $\pm$ 12.2	65.6 $\pm$ 11.6	0.005*
Male	80 (40.8)	26 (30.2)	0.120†
IOP (mm Hg)	13.7 $\pm$ 5.3	15.0 $\pm$ 3.0	0.009*
SE (D)	-1.00 $\pm$ 2.99	-1.07 $\pm$ 2.67	0.878*
GCIPL thickness (μ m)			
Average	81.1 $\pm$ 6.2	66.7 $\pm$ 7.4	< 0.001*
Minimum	77.7 $\pm$ 7.9	54.0 $\pm$ 11.5	< 0.001*
Superotemporal	79.9 $\pm$ 8.0	68.0 $\pm$ 8.4	< 0.001*
Inferotemporal	80.7 $\pm$ 7.8	59.9 $\pm$ 10.4	< 0.001*
Inferior	79.1 $\pm$ 6.9	60.4 $\pm$ 9.6	< 0.001*
Inferonasal	81.3 $\pm$ 6.3	67.5 $\pm$ 11.1	< 0.001*
Superonasal	83.5 $\pm$ 6.8	73.9 $\pm$ 10.9	< 0.001*
Superior	82.0 $\pm$ 7.1	70.9 $\pm$ 10.5	< 0.001*
RNFL thickness (μ m)			
Average	92.5 $\pm$ 9.1	69.2 $\pm$ 11.1	< 0.001*
Superior	118.6 $\pm$ 44.6	86.6 $\pm$ 19.5	< 0.001*
Temporal	68.7 $\pm$ 12.1	55.9 $\pm$ 12.9	< 0.001*
Inferior	118.5 $\pm$ 14.3	74.9 $\pm$ 19.9	< 0.001*
Nasal	67.1 $\pm$ 8.8	59.4 $\pm$ 9.6	< 0.001*
SAP C24-2 MD (dB)	-0.47 $\pm$ 2.11	-6.45 $\pm$ 6.02	< 0.001*

Data are mean $\pm$ SD or n (%) unless otherwise indicated.

*Student *t* test.

† χ^2 test.

GCIPL indicates ganglion cell-inner plexiform layer; IOP, intraocular pressure; MD, mean deviation; RNFL, retinal nerve fiber layer; SAP, standard automated perimetry; SE, spherical equivalent.

parameters, inferior RNFL thickness showed the highest AUC (0.946; 95% CI, 0.918-0.975). The receiver operating characteristics curve of the ensemble training model, minimum GCIPL thickness, and inferior RNFL thickness are drawn in Figure 3.

Subgroup Analysis for Early and Moderate-to-Severe Glaucoma

To evaluate the diagnostic performance according to severity, the glaucoma group in the test set was divided into early (SAP MD > -6.00 dB) and moderate-to-severe glaucoma (SAP MD $\leq$ -6.00 dB), and subsequent analysis of AUC was performed. For the test dataset, 14 eyes were included in the early glaucoma group and 12 eyes were included in the moderate-to-severe glaucoma group. As expected, the AUCs of the overall analysis was higher than that for discrimination of early glaucoma and lower than that for differentiation of moderate-to-severe glaucoma in most clinical parameters (Table 4). Nevertheless, the AUC of the ensemble training model was still excellent for the detection of early glaucoma (0.980; 95% CI, 0.952-1.000) and higher than those of most of the conventional clinical parameters, except for minimum GCIPL thickness (*P* = 0.056). Furthermore, the ensemble training model successfully differentiated all of the moderate-to-severe glaucoma cases from the normal cases (AUC 1.000; 95% CI, 0.998-1.000); however, that AUC was not significantly higher than all clinical parameters. The AUCs of the minimum, inferior, and inferotemporal GCIPL thicknesses, as well as the average and inferior RNFL, did not significantly differ from that of the ensemble training model (*P* = 0.067, 0.235, 0.130, 0.145, and 0.121, respectively).

TABLE 2. Areas Under Receiver Operating Characteristic Curve of Deep Learning Inference Models and Conventional Diagnostic Parameters on Test Sets

	AUC (Range)	P*	Sensitivity [% (Range)]	Specificity [% (Range)]
Ensemble training model	0.990 (0.975-1.000)	NA	94.7 (90.3-98.2)	100 (100-100)
Deep learning model with single modality				
GCIPL thickness map	0.981 (0.967-0.996)	0.092	92.2 (87.0-96.5)	96.9 (92.6-100)
GCIPL deviation map	0.986 (0.976-0.997)	0.436	93.9 (88.7-98.3)	95.8 (91.6-99.0)
RNFL thickness map	0.979 (0.963-0.995)	0.091	96.5 (93.0-99.1)	91.6 (85.3-96.9)
RNFL deviation map	0.962 (0.935-0.989)	0.026	91.3 (86.1-96.5)	93.7 (88.4-86.1)
GCIPL thickness				
Average	0.949 (0.921-0.976)	0.006	90.5 (84.6-94.8)	91.3 (86.1-95.9)
Minimum	0.963 (0.934-0.985)	0.026	86.0 (78.9-92.1)	98.8 (96.5-100)
Superotemporal	0.914 (0.875-0.953)	<0.001	80.0 (72.2-87.0)	94.7 (89.5-99.0)
Superior	0.826 (0.770-0.881)	<0.001	61.7 (53.0-70.4)	92.6 (87.4-96.8)
Superonasal	0.777 (0.713-0.840)	<0.001	60.0 (51.3-68.7)	88.4 (82.1-94.7)
Inferonasal	0.877 (0.830-0.925)	<0.001	75.7 (67.8-82.6)	82.1 (74.7-87.5)
Inferior	0.951 (0.923-0.979)	0.007	91.3 (85.2-95.7)	89.5 (83.2-94.7)
Inferotemporal	0.954 (0.926-0.981)	0.012	85.2 (78.3-91.3)	98.9 (96.8-100)
RNFL thickness				
Average	0.938 (0.905-0.971)	0.003	93.0 (87.8-97.4)	86.3 (79.0-92.6)
Superior	0.872 (0.824-0.921)	<0.001	92.2 (86.9-96.5)	68.4 (57.9-77.9)
Nasal	0.750 (0.684-0.816)	<0.001	56.5 (47.8-65.2)	88.4 (81.1-94.7)
Inferior	0.946 (0.918-0.975)	0.004	91.3 (86.1-95.7)	84.2 (76.8-91.6)
Temporal	0.798 (0.738-0.858)	<0.001	80.9 (73.0-87.9)	68.4 (57.9-76.8)
SAP C24-2 MD	0.889 (0.844-0.934)	<0.001	82.5 (75.4-89.5)	82.9 (74.4-91.5)

*Comparison with ensemble training model (DeLong test).

AUC indicates area under the curve; GCIPL, ganglion cell-inner plexiform layer; MD, mean deviation; NA, not applicable; RNFL, retinal nerve fiber layer; SAP, standard automated perimetry.

Analysis of False Positivity (FP) and False Negativity (FN)

To identify the cause of misclassification and, thus, improve the diagnostic algorithm, all of the FP and FN cases were comprehensively reviewed. With the cutoff probability having been set at 0.5, there were only 2 FP cases and 1 FN case. Figure 4A shows a representative FP

case of a 69-year-old woman. There were no definite RNFL defects or glaucomatous visual-field defects on the pattern deviation plot; thus, this case should have been classified as “normal.” However, the ensemble model predicted

TABLE 3. P-values of Comparison of Areas Under the Curve Between Deep Learning Models With Single Modality and Conventional Diagnostic Parameters by DeLong Test

	Deep Learning Models With Single Modality			
	GCIPL Thickness Map	GCIPL Deviation Map	RNFL Thickness Map	RNFL Deviation Map
GCIPL thickness				
Average	0.019	0.006	0.067	0.540
Minimum	0.139	0.040	0.215	0.964
Superotemporal	<0.001	<0.001	0.002	0.019
Superior	<0.001	<0.001	<0.001	<0.001
Superonasal	<0.001	<0.001	<0.001	<0.001
Inferonasal	<0.001	<0.001	<0.001	<0.001
Inferior	0.020	0.007	0.071	0.544
Inferotemporal	0.031	0.010	0.093	0.643
RNFL thickness				
Average	0.021	0.007	0.015	0.162
Superior	<0.001	<0.001	<0.001	<0.001
Nasal	<0.001	<0.001	<0.001	<0.001
Inferior	0.026	0.005	0.031	0.349
Temporal	<0.001	<0.001	<0.001	<0.001
SAP C24-2 MD	<0.001	<0.001	<0.001	0.006

GCIPL indicates ganglion cell-inner plexiform layer; MD, mean deviation; RNFL, retinal nerve fiber layer; SAP, standard automated perimetry.

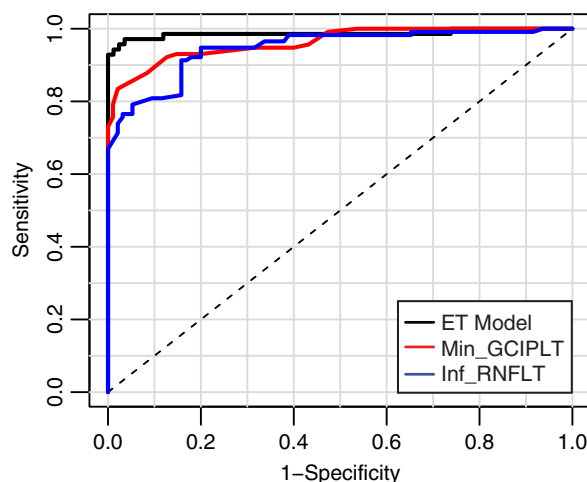
**FIGURE 3.** Receiver operating characteristic curves comparing diagnostic performance between the ensemble training model, the minimum GCIPL, and the inferior sectoral RNFL thickness parameters using the test dataset. The area under the curve of the ensemble training model was 0.990, which was significantly higher than those of the inferior RNFL and minimum GCIPL thickness parameters ($P=0.004$ and 0.026 , respectively). ET indicates ensemble training; GCIPL, ganglion cell-inner plexiform layer; Inf_RNFLT, inferior retinal nerve fiber layer thickness; Min_GCIPLT, minimum ganglion cell-inner plexiform layer thickness; NAS, neural architecture search; RNFL, retinal nerve fiber layer. Figure 3 can be viewed in color online at www.glaucomajournal.com.

TABLE 4. Areas Under Receiver Operating Characteristic Curve of Deep Learning Prediction Models for Differentiation of Early and Moderate-to-Severe Glaucoma From Healthy Controls on Test Set

	Early Glaucoma (N = 14)		Moderate-to-Severe Glaucoma (N = 12)	
	AUC (Range)	P	AUC (Range)	P
Ensemble training model	0.980 (0.952-1.000)	NA	1.000 (0.998-1.000)	NA
GCIPL thickness				
Average	0.897 (0.841-0.953)	0.007	0.990 (0.980-0.998)	0.048
Minimum	0.931 (0.890-0.973)	0.056	0.992 (0.983-1.000)	0.067
Superotemporal	0.876 (0.813-0.939)	0.001	0.952 (0.912-0.992)	0.019
Superior	0.727 (0.640-0.814)	< 0.001	0.921 (0.868-0.974)	0.004
Superonasal	0.693 (0.597-0.788)	< 0.001	0.860 (0.788-0.932)	< 0.001
Inferonasal	0.808 (0.729-0.887)	< 0.001	0.940 (0.94-0.975)	< 0.001
Inferior	0.901 (0.845-0.956)	0.004	0.998 (0.996-1.000)	0.235
Inferotemporal	0.917 (0.865-0.969)	0.026	0.990 (0.978-1.000)	0.130
RNFL thickness				
Average	0.893 (0.833-0.952)	0.009	0.980 (0.953-1.000)	0.145
Superior	0.817 (0.746-0.889)	< 0.001	0.938 (0.796-0.980)	0.005
Nasal	0.668 (0.572-0.764)	< 0.001	0.823 (0.746-0.900)	< 0.001
Inferior	0.899 (0.846-0.951)	0.005	0.993 (0.984-1.000)	0.121
Temporal	0.733 (0.649-0.818)	< 0.001	0.863 (0.795-0.932)	< 0.001
SAP C24-2 MD	0.795 (0.719-0.872)	< 0.001	1.000 (0.999-1.000)	0.940

AUC indicates area under curve; GCIPL, ganglion cell-inner plexiform layer; MD, mean deviation; NA, not applicable; RNFL, retinal nerve fiber layer; SAP, standard automated perimetry.

glaucoma with a probability of 72.8%. Notably, a Weiss' ring was observed in the red-free fundus photograph, which made a small artifact on the RNFL thickness and deviation maps, and which was the likely cause of the misclassification. In another FP case of a 51-year-old woman with high myopia (−8.75 D), the peripapillary atrophy had affected the classification process for the RNFL OCT. Figure 4B shows the FN case of a 67-year-old man. The red-free fundus photograph revealed inferior diffuse RNFL atrophy, which corresponded to a superior visual field defect. Interestingly, the deviation maps were classified as glaucoma (GCIPL 63.2%, RNFL 99.0%) whereas the thickness maps were classified as normal (GCIPL 23.3%, RNFL 11.8%). The ensemble training model predicted the glaucoma probability of 17.4%. Atrophy of the retinal pigment epithelium seemed to have disturbed the decision process of our algorithm.

DISCUSSION

In this study, the deep learning classifier with ensemble training differentiated glaucomatous change from healthy controls with robust performance. Even the deep learning models trained with only a single modality showed very high performance.

Muhammad et al¹⁴ introduced a hybrid deep learning method with a single wide-field OCT protocol. They integrated AlexNet and the random forest model to achieve a promising result for discrimination of glaucomatous change. However, only single imaging modalities (ie, RNFL thickness map or RNFL probability map) were used as input in their study. To overcome this weakness, they combined GCIPL and RNFL probability images; however, this did not yield superior results to those of RNFL probability alone. They suggested that a novel method of combining RNFL and GCIPL information would be a feasible way to enhance diagnostic performance. Also, the recent study by Asaoka et al²³ revealed that deep learning and transform learning can effectively detect the early-onset glaucomatous

change in macular OCT images. The deep learning model showed significantly larger AUC than classical machine learning methods, such as random forest or support vector machine. However, they only used macular OCT images, not also RNFL OCT images. Recently, a transfer-learning model using combined feature vectors of fundus photography and SD-OCT image maps was introduced for glaucoma diagnosis.²⁴ Similarly, we adopted the NASNet architecture to extract important bottleneck features of 4 images that had been entered together into an additional fully-connected deep neural network. Thereby, the deep learning model successfully discriminated the glaucomatous eyes from healthy eyes.

Our study has some advantages relative to previous work. First, we used 4 images simultaneously to train the deep learning model. Since each of the OCT images has its unique merits for the diagnosis of glaucoma, it is crucial to consider all of them in real-world practice. Likewise, a deep learning model trained with all of the OCT maps would be expected to offer better performance. In our study, although a significant difference was found only between the ensemble training model and the RNFL deviation map model, the AUC of the ensemble training model was higher than those of the single-modality models. Second, we included a class-balanced dataset for glaucoma and normal groups. Unbalanced recruitment between 2 groups would result in biased training and underwhelming performance for the test dataset. In our dataset, the numbers of total OCT image sets for the 2 groups were similar [307 (46.7%) vs. 350 (53.3%), respectively]. In this manner, we could avoid the problems of biased training and estimation. Third, we adopted NASNet, a state-of-the-art deep neural network that showed excellent accuracy, as a baseline deep learning architecture to extract important bottleneck features. Moreover, the expected input shape of NASNet (331, 331, 3) is larger than those of most other neural networks, such as Inception-v3's (224, 224, 3). Considering that it is important to preserve as much information as possible for the training of the deep learning model, the choice of NASNet in this study was justified.

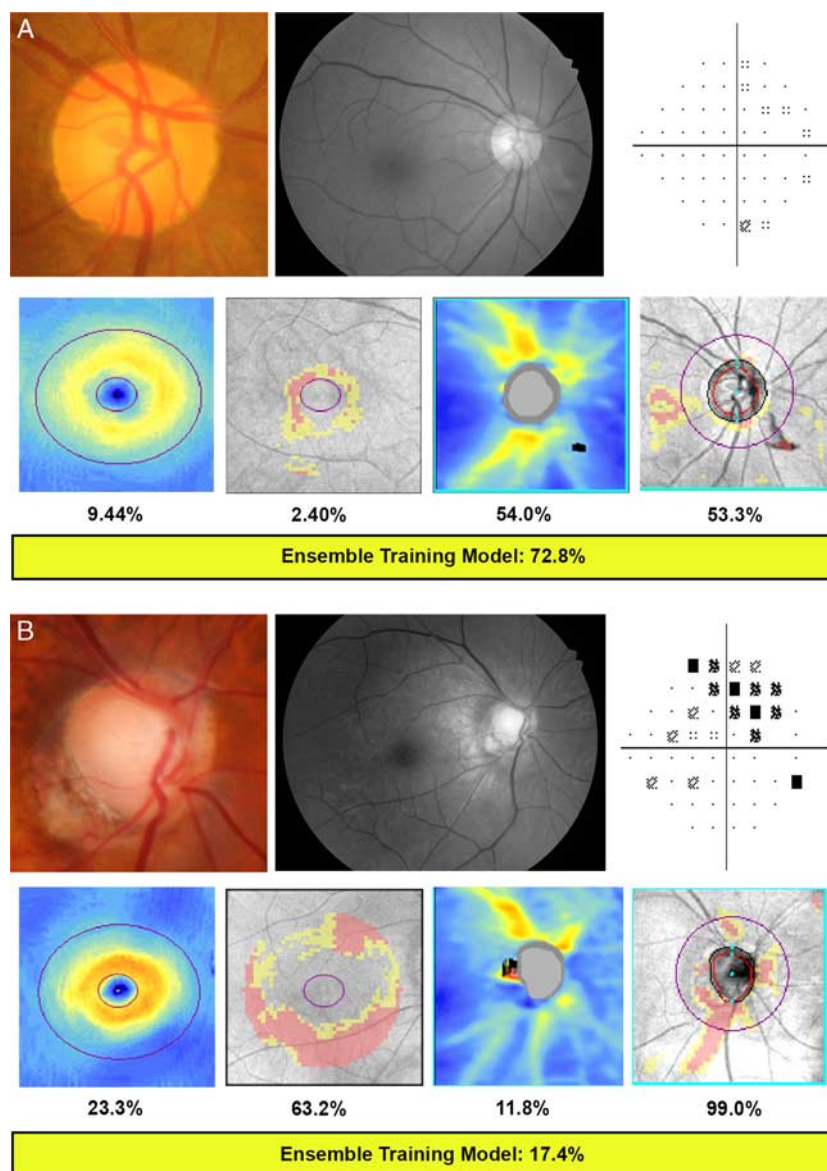


FIGURE 4. Representative cases of deep learning algorithm misclassification. A, False-positive (FP) case of a 69-year-old woman. No definite retinal nerve fiber layer (RNFL) defect was shown on red-free RNFL photography, and the pattern deviation plot likewise showed no glaucomatous visual-field change. However, the ensemble model predicted glaucoma with a probability of 72.8%. Weiss' ring was observed in the red-free fundus photograph and made a small artifact on the RNFL thickness and deviation maps, which was the likely cause of misclassification. B, False-negative (FN) case of a 67-year-old man. Red-free RNFL photography and the RNFL optical coherence tomography deviation map demonstrate inferior diffuse RNFL atrophy corresponding to a superior visual-field defect on the pattern deviation plot. Each deviation map was classified as glaucoma [ganglion cell-inner plexiform layer (GCIPL) 63.2%, RNFL 99.0%], while the thickness maps were classified as normal (GCIPL 23.3%, RNFL 11.8%). The ensemble training model predicted the glaucoma probability of 17.4%. Atrophy of the retinal pigment epithelium was evident, which might have disturbed the decision process of our algorithm. Figure 4 can be viewed in color online at www.glaucomajournal.com.

It is known that nearly half of FN cases and about 19% of FP cases of glaucoma are associated with optic disc change secondary to high myopia or pathologic myopia.²⁵ In our test dataset, an FP case also presented high myopia. Further study to reduce the error rate in high myopia or pathologic myopia is expected. Another interesting aspect associated with FP cases is the Weiss ring, which was not expected to make for misclassification when we developed the program. Analysis of vitreoretinal findings that jeopardize diagnostic performance certainly would be valuable research.

There are some limitations to our study. First, the number of datasets was relatively small and only highly selected normal-tension glaucoma patients were included. Considering that healthy controls have IOP within the normal range, we considered that normal-tension glaucoma might be more difficult to discriminate than high-tension glaucoma in clinical practice. However, further study with more datasets including high-tension glaucoma, for generalization purposes, would be warranted. Second, there was no clarity on how the deep learning model classified cases as

glaucoma or normal. This, the so-called “black box problem,” is a well-known weak point of the deep learning algorithm. There have been many attempts to visualize what a deep learning network learns, which process will be applied in a future clinical study. Third, this model was developed only to discriminate between glaucomatous optic neuropathy and healthy controls. However, there are many nonglaucomatous optic neuropathies presenting RNFL defects on OCT images. In our model, it is possible that images of nonglaucomatous optic neuropathy are misdiagnosed as glaucoma. Fourth, the diagnosis of glaucoma was made based on only stereo disc photography and red-free fundus photography, regardless of visual-field defects. Since we trained and validated the deep learning classifier utilizing OCT images, which were also structural data, this might have biased the diagnostic ability by overestimating the sensitivity-specificity balance. Moreover, we excluded the eyes from the analysis when the structural judgment of glaucoma by clinicians differed. That is, the algorithm was trained and tested by relatively obvious cases, which could have led to accuracy overestimation. Finally, although NASNet is an outstanding deep neural network, it was not designed for recognition of OCT images. It would be interesting to design and train a convolutional neural network specialized for OCT images.

In conclusion, the deep learning model outperformed the standard OCT and SAP clinical parameters in discriminating glaucomatous eyes from healthy eyes. It might be possible to further improve its performance by increasing the size of the training set and creating heat maps for visualization of what the deep neural network has learned.

REFERENCES

1. Dandona L, Dandona R. What is the global burden of visual impairment? *BMC Med*. 2006;4:6.
2. Foster PJ, Buhrmann R, Quigley HA, et al. The definition and classification of glaucoma in prevalence surveys. *Br J Ophthalmol*. 2002;86:238–242.
3. Leske MC. The epidemiology of open-angle glaucoma: a review. *Am J Epidemiol*. 1983;118:166–191.
4. Leske MC, Heijl A, Hyman L, et al. Early Manifest Glaucoma Trial: design and baseline data. *Ophthalmology*. 1999;106:2144–2153.
5. Anderson DR. Collaborative normal tension glaucoma study. *Curr Opin Ophthalmol*. 2003;14:86–90.
6. Bowd C, Weinreb RN, Williams JM, et al. The retinal nerve fiber layer thickness in ocular hypertensive, normal, and glaucomatous eyes with optical coherence tomography. *Arch Ophthalmol*. 2000;118:22–26.
7. Choi YJ, Jeoung JW, Park KH, et al. Glaucoma detection ability of ganglion cell-inner plexiform layer thickness by spectral-domain optical coherence tomography in high myopia. *Invest Ophthalmol Vis Sci*. 2013;54:2296–2304.
8. Hwang YH, Jeoung YC, Kim HK, et al. Macular ganglion cell analysis for early detection of glaucoma. *Ophthalmology*. 2014;121:1508–1515.
9. Jeoung JW, Choi YJ, Park KH, et al. Macular ganglion cell imaging study: glaucoma diagnostic accuracy of spectral-domain optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2013;54:4422–4429.
10. Kim KE, Park KH. Macular imaging by optical coherence tomography in the diagnosis and management of glaucoma. *Br J Ophthalmol*. 2018;102:718–724.
11. Gulshan V, Peng L, Coram M, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*. 2016;316:2402–2410.
12. Ting DSW, Cheung CYL, Lim G, et al. Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes. *JAMA*. 2017;318:2211–2223.
13. Chen X, Xu Y, Wong DWK, et al. Glaucoma detection based on deep convolutional neural network. Engineering in Medicine and Biology Society (EMBC), 2015 37th Annual International Conference of the IEEE: IEEE; 2015.
14. Muhammad H, Fuchs TJ, De Cuir N, et al. Hybrid deep learning on single wide-field optical coherence tomography scans accurately classifies glaucoma suspects. *J Glaucoma*. 2017;26:1086–1094.
15. Fleming AD, Philip S, Goatman KA, et al. Automated assessment of diabetic retinal image quality based on clarity and field definition. *Invest Ophthalmol Vis Sci*. 2006;47:1120–1125.
16. Zoph B, Le QV. Neural architecture search with reinforcement learning. 2016. Available at: <https://arxiv.org/abs/1611.01578>. Accessed February 9, 2020.
17. Zoph B, Vasudevan V, Shlens J, et al. Learning transferable architectures for scalable image recognition. Proceedings of the IEEE conference on computer vision and pattern recognition; 2018.
18. Szegedy C, Vanhoucke V, Ioffe S, et al. Rethinking the inception architecture for computer vision. Proceedings of the IEEE conference on computer vision and pattern recognition; 2016.
19. He K, Zhang X, Ren S, et al. Deep residual learning for image recognition. Proceedings of the IEEE conference on computer vision and pattern recognition; 2016.
20. Japkowicz N, Shah M. *Evaluating learning algorithms: a classification perspective*. City of Cambridge: Cambridge University Press; 2011.
21. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44:837–845.
22. Team RC. R: a language and environment for statistical computing; 2013.
23. Asaoka R, Murata H, Hirasawa K, et al. Using deep learning and transform learning to accurately diagnose early-onset glaucoma from macular optical coherence tomography images. *Am J Ophthalmol*. 2019;198:136–145.
24. An G, Omodaka K, Hashimoto K, et al. Glaucoma diagnosis with machine learning based on optical coherence tomography and color fundus images. *J Healthc Eng*. 2019;2019:4061313.
25. Li Z, He Y, Keel S, et al. Efficacy of a deep learning system for detecting glaucomatous optic neuropathy based on color fundus photographs. *Ophthalmology*. 2018;125:1199–1206.